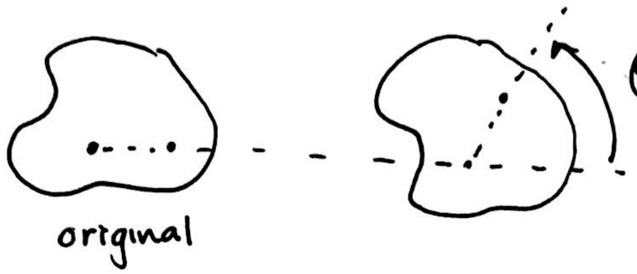
MAE 4730
Lecture 17
10/3/2018

Today:
① 2D Kinematics
② Rigid objects (2D) review
③ Inverted pendulum shaking base

2D Rigid Object Kinematics



Minimal coordinates:

$\vec{r}_{G/O}$, θ ← rotation

position ← reference point
can be COM

$R = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$ can also use rotation matrix

Angular Velocity: 2D: $\boxed{\omega = \dot{\theta}\hat{k}}$ — ω is a property of the object
- Different θ, but same $\dot{\theta}$

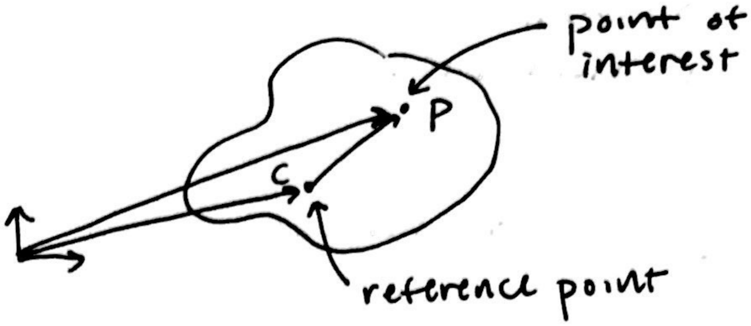
Key equation: $\boxed{\vec{V}_{2/1} = \vec{\omega} \times \vec{r}_{2/1}}$

- For any 2 pts in rigid object, point is going in circles around the other one.

Angular Acceleration: $\vec{\alpha} = \frac{d}{dt}\vec{\omega}$

$\boxed{2D: \vec{\alpha} = \ddot{\theta}\hat{k}}$

want position, velocity + acceleration of pts on a rigid object



pos:

$$\boxed{\vec{r}_p = \vec{r}_c + \vec{r}_{p/c}}$$

$\vec{r}_{p/o}$

vel:

$$\vec{v}_p = \dot{\vec{r}}_c + \frac{d}{dt}(\vec{r}_{p/c})$$

$$\boxed{\vec{v}_p = \vec{v}_c + \vec{\omega} \times \vec{r}_{p/c}}$$

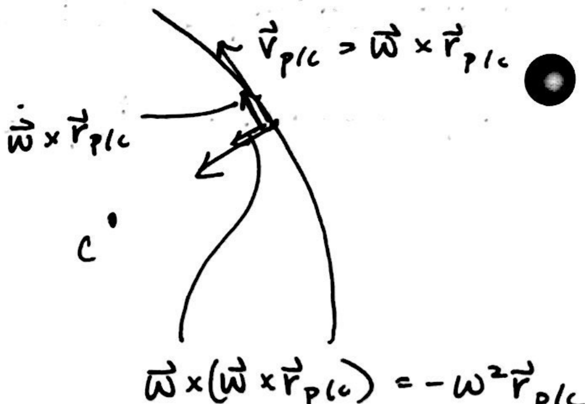
accel:

$$\vec{a}_p = \frac{d}{dt}(\vec{v}_p)$$

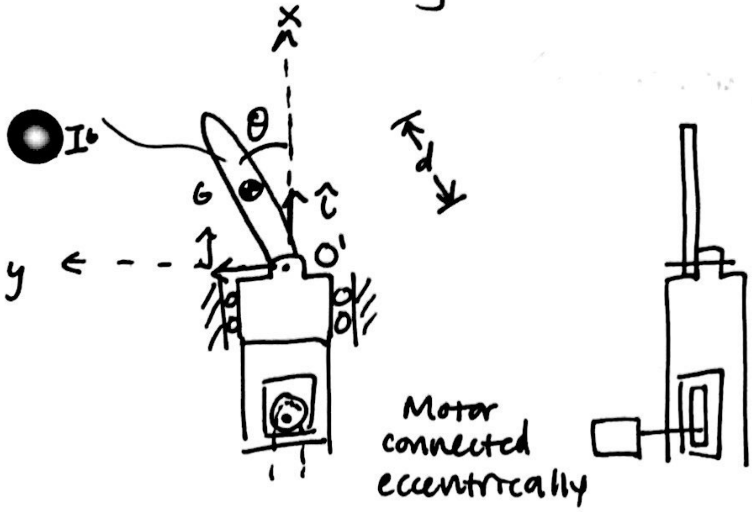
$$= \frac{d}{dt}\left[\vec{v}_c + \vec{\omega} \times \vec{r}_{p/c}\right]$$

$$= \vec{a}_c + \dot{\vec{\omega}} \times \vec{r}_{p/c} + \vec{\omega} \times \dot{\vec{r}}_{p/c}$$

$$\boxed{\vec{a}_p = \vec{a}_c + \vec{\alpha} \times \vec{r}_{p/c} + \underbrace{\vec{\omega} \times (\vec{\omega} \times \vec{r}_{p/c})}_{\text{in 2D: } = -\omega^2 \vec{r}_{p/c}}}$$



ex) Inverted Shaking Pendulum



Fixed origin O under O' in picture

2D:

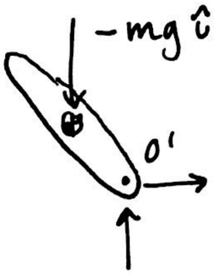
· Given: d, m, I^G, O' motion: $\boxed{\vec r_{O'/O} = h\sin(\omega t)\,\hat\imath}$

· Find EoM:

Method 1 | N-E w/ minimal coordinates

1 DoF $\Rightarrow \theta$

FBD:



AMB$_{/O'}$: $\boxed{\sum \vec M^{ext}_{/O'} = \vec r_{G/O} \times m\vec a_G + I^G \alpha\,\hat k}$ ✓

an accelerating point

$\neq \frac{d}{dt}\left(\vec H_{/O'}\right)$ ✗

$$\vec r_{G/O'} \times (-mg\,\hat\imath) = \vec r_{G/O'} \times m\vec a_G + I^G \ddot\theta\,\hat k$$

$d\hat e_r$

$\cos\theta\,\hat\imath + \sin\theta\,\hat\jmath$

$\vec a_{O'} + \vec a_{G/O'}$

$\ddot\theta\,\hat k \times \vec r_{G/O'} + -\dot\theta^2\,\vec r_{G/O'}$

$-\omega^2 h \sin(\omega t)\,\hat\imath$

Do all cross products $\Rightarrow$ E.O.M.

$\ddot{\theta}$ = something involving $(\theta, \dot{\theta}, t,$ parameters$)$